

MICROSOFT TEAMS REPLICA

Project Documentation

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1. Introduction

The Microsoft Teams Replica is a web-based collaboration platform designed to provide communication, teamwork, and productivity tools within a single application. The platform enables users to communicate through chat, audio calls, video meetings, file sharing, task management, and team collaboration features.

The system helps organizations, educational institutions, and remote teams work efficiently by providing a centralized workspace where users can communicate, share information, schedule meetings, and collaborate on projects.

2. Problem Statement

Modern organizations rely on multiple applications for communication, meetings, file sharing, task management, and collaboration. Using different tools creates communication gaps, data fragmentation, reduced productivity, and management difficulties.

Employees often struggle to manage conversations, meetings, files, and tasks across multiple platforms. Organizations require a centralized communication and collaboration system that integrates messaging, meetings, document sharing, task management, and team collaboration into a single platform.

Therefore, a Microsoft Teams Replica is required to improve communication, collaboration, productivity, and organizational efficiency.

3. Proposed Solution

The proposed Microsoft Teams Replica provides a centralized platform for communication and collaboration. Users can exchange messages, conduct audio and video meetings, share files, manage tasks, collaborate on documents, and communicate within teams and channels.

The platform combines communication and productivity features into one application, reducing the need for multiple tools while improving teamwork and workflow management.

4. Objectives

- Improve communication among team members.
 - Centralize collaboration activities.
 - Simplify meeting management.
 - Enable secure file sharing.
 - Improve task tracking and productivity.
 - Support remote and hybrid work environments.
 - Provide real-time collaboration features.
 - Enhance organizational efficiency.
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5. System Modules

User Management Module

Functions

- User Registration
 - Login
 - Profile Management
 - Department Management
 - User Roles
-

Personal Chat Module

Functions

- One-to-One Chat
 - Media Sharing
 - Message Search
 - Read Receipts
 - Reactions
-

Group Chat Module

Functions

- Group Creation
 - Member Management
 - Shared Files
 - Pinned Messages
-

Teams & Channels Module

Functions

- Team Creation
 - Channel Management
 - Team Announcements
 - Member Invitations
-

Audio & Video Communication Module

Functions

- Audio Calls
 - Video Meetings
 - Meeting Scheduling
 - Screen Sharing
 - Recording
-

File Management Module

Functions

- Upload Files
- Share Files
- Download Files

- File Organization
-

Task Management Module

Functions

- Task Assignment
 - Deadline Tracking
 - Progress Monitoring
 - Team Tasks
-

Administration Module

Functions

- User Management
 - Team Management
 - Security Controls
 - Reports & Analytics
-

6. Implementation Details

User Authentication Module

This module manages user registration and login functionality.

Features

- Registration
- Login
- Password Recovery
- Role-Based Access Control

Implementation

User information is stored in MySQL. JWT authentication is used to maintain secure user sessions and role-based access permissions.

Personal & Group Chat Module

This module enables communication between users.

Features

- One-to-One Messaging
- Group Messaging
- Media Sharing
- Message Search

Implementation

Users can send messages in real-time. Messages are stored in the database and synchronized using Socket.IO.

Teams & Channels Module

This module organizes users into teams and communication channels.

Features

- Create Teams
- Create Channels
- Invite Members
- Team Discussions

Implementation

Teams and channels are created by administrators or team owners. Members can participate in conversations and access shared resources.

Audio & Video Meeting Module

This module supports virtual communication.

Features

- Audio Calls
- Video Conferencing
- Screen Sharing
- Meeting Scheduling

Implementation

WebRTC is used to establish peer-to-peer communication for audio and video calls. Meeting information is maintained in the database.

File Management Module

This module handles file storage and sharing.

Features

- File Upload
- File Download
- File Sharing
- File Preview

Implementation

Users can upload and share documents. File information is stored and linked to chats, teams, and channels.

Task Management Module

This module helps teams organize work.

Features

- Create Tasks
- Assign Tasks
- Set Deadlines
- Track Progress

Implementation

Managers assign tasks to team members. Task completion status is updated and monitored through dashboards.

Calendar Module

This module manages schedules and meetings.

Features

- Daily Calendar
- Weekly Calendar
- Meeting Reminders
- Event Scheduling

Implementation

Users can schedule meetings and receive reminders for upcoming events.

Notification Module

This module keeps users informed about activities.

Features

- Message Notifications
- Meeting Alerts
- Task Notifications

Implementation

Notifications are generated automatically whenever important actions occur within the platform.

Administration Module

This module manages overall system operations.

Features

- User Management
- Team Management
- Analytics
- Security Settings

Implementation

Administrators control user permissions, team structures, and system settings.

7. System Workflow

1. User registers and logs into the platform.
 2. Teams and channels are created.
 3. Users join teams and communicate through chats.
 4. Meetings are scheduled and conducted.
 5. Files are uploaded and shared.
 6. Tasks are assigned and monitored.
 7. Notifications are generated automatically.
 8. Reports and analytics are generated for administrators.
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8. Technologies Used

Component	Technology
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	Node.js, Express.js
Database	MySQL
Authentication	JWT

Component	Technology
Real-Time Communication	Socket.IO
Video Communication	WebRTC
IDE	Visual Studio Code
Version Control	GitHub
Deployment	Vercel / Render

9. Expected Outcomes

- Improved team communication.
 - Better collaboration among employees.
 - Efficient meeting management.
 - Centralized file sharing.
 - Improved task tracking.
 - Enhanced productivity.
 - Secure communication platform.
 - Better organizational workflow.
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10. Future Enhancements

- AI Meeting Summaries.
 - Smart Task Recommendations.
 - Mobile Application.
 - Collaborative Whiteboard.
 - Polls & Surveys.
 - Webinar Support.
 - Advanced Analytics Dashboard.
 - Cloud Storage Integration.
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11. Conclusion

The Microsoft Teams Replica is a comprehensive communication and collaboration platform that integrates messaging, meetings, file sharing, task management, and team collaboration into a single system. The platform improves communication efficiency, enhances productivity, and supports modern workplace requirements.

By providing a centralized workspace for collaboration, the system helps organizations streamline workflows, improve teamwork, and support remote and hybrid working environments.